Chemical Kinetics

ChemieLearn

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Defining Rate of Reactions

The average rate of a chemical reaction is defined as the change in concentration of a reactant or product over a specific time interval divided by that time interval. Mathematically, it can be expressed as:

$$\mathrm{A} \longrightarrow \mathrm{B} \Longrightarrow \boxed{r = -\frac{\Delta[\mathrm{A}]}{\Delta t} = \frac{\Delta[\mathrm{B}]}{\Delta t}}$$

where $\Delta[\mathrm{A}]$ or $\Delta[\mathrm{B}]$ represents the change in concentration of reactant A or product B, respectively, and Δt represents the corresponding change in time. Assuming that the reaction coefficient of reactant A is denoted as c_a, and the reaction coefficient for product B is denoted as c_b, the average rate of the reaction can be expressed as follows:

$$c_a\mathrm{A} \longrightarrow c_b\mathrm{B} \Longrightarrow \boxed{r = -\frac{1}{c_a} \cdot \frac{\Delta[\mathrm{A}]}{\Delta t} = \frac{1}{c_b} \cdot \frac{\Delta[\mathrm{B}]}{\Delta t}}$$

The instantaneous rate of a chemical reaction refers to the rate of change of the concentration of a reactant or product at a specific point in time. It is obtained by taking the derivative of the concentration with respect to time. Mathematically, the instantaneous rate is given by:

$$\begin{aligned} r_t &= \lim_{\Delta t \to 0} \left(-\frac{1}{\alpha} \cdot \frac{[\mathrm{A}]_{t+\Delta t} - [\mathrm{A}]_t}{\Delta t} \right) \\ &= -\frac{1}{\alpha} \cdot \frac{\mathrm{d}[\mathrm{A}]}{\mathrm{d}t} \end{aligned}$$

Measuring Rate of Reactions

In order to determine the rate of a chemical reaction, it is necessary to measure the changes in concentration over a specific period of time. While the measurement of time can be accomplished using a stopwatch or other timing device, the measurement of concentration changes during a chemical reaction requires appropriate analytical techniques. The term “average” is used when referring to the rate of a reaction because it represents an overall measure of the reaction's progress over a given time interval. The average rate is calculated by dividing the change in concentration by the corresponding change in time. It provides a general indication of how the concentrations of reactants and products evolve during the reaction.

However, it is important to note that the rate of a reaction is not constant throughout the entire process. It may vary at different time points due to factors such as changes in reactant concentrations, temperature, and the influence of catalysts. The instantaneous rate, which represents the rate of

change at a specific moment in time, provides a more detailed and precise understanding of the reaction kinetics.

Experiment

A 3% aqueous solution of hydrogen peroxide is widely employed as an antiseptic. The antiseptic action is attributed to the release of oxygen gas (O_2) as hydrogen peroxide undergoes decomposition. The oxygen gas escapes from the solution, and ultimately, the decomposition reaction reaches completion.

$$H_2O_2(aq) \longrightarrow H_2O(l) + \frac{1}{2}O_2(g)$$

The progress of the reaction can be tracked by focusing on either the formation of oxygen gas (O_2) or the disappearance of hydrogen peroxide (H_2O_2). There are various techniques available to monitor and analyze these changes in concentrations over time.

One approach involves quantifying the volumes of oxygen gas produced at different time intervals and establishing a relationship between these volumes and the corresponding decreases in hydrogen peroxide concentration. This data can be graphed or plotted to visualize the reaction's progression, as depicted in Figure 1.

Another method entails extracting small samples from the reaction mixture at specific time points and subjecting these samples to chemical analysis. One common technique is titration, which can be performed using an acidic solution and a reducing agent such as potassium permanganate ($KMnO_4$). The net ionic equation for this oxidation-reduction reaction can be written to represent the electron transfer between hydrogen peroxide and potassium permanganate.

$$2MnO_4^-(aq) + 5H_2O_2(aq) + 6H^+(aq) \longrightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5O_2(g)$$

Data Extrapolation

When the rate of a reaction is expressed as an average value, it represents the average rate over a specific time interval. For instance, consider the interval from 1200 to 1600 seconds. The average rate for this interval can be determined, resulting in a value such as the fourth entry in the fifth column of Table 1. This average rate can be seen as the rate of the reaction at approximately the midpoint of the interval, around 1400 seconds. However, it is important to note that the choice of interval boundaries is arbitrary, and different intervals can yield slightly different average rates.

In Table 1, alternative choices of interval boundaries could have been made. For example, one could have selected a different interval, such as from 1300 to 1500 seconds. In this case, to calculate the rate of reaction at 1400 seconds, concentration data at both 1300 and 1500 seconds would be utilized. The resulting rate of reaction would be slightly different from the previously mentioned average rate.

As the time interval used becomes smaller, the calculated average rate approaches the actual rate at a specific moment. The instantaneous rate of reaction at a specific point on a concentration-time curve is obtained by calculating the slope of the tangent line that touches the curve at that point. By choosing intervals with narrower time spans, the average rate becomes a more accurate

representation of the instantaneous rate. As the time interval, represented by Δt, approaches zero seconds ($\Delta t \to 0$), the rate of reaction at $t = 1400$ seconds becomes the instantaneous rate, and the value of Δt becomes infinitesimally small.

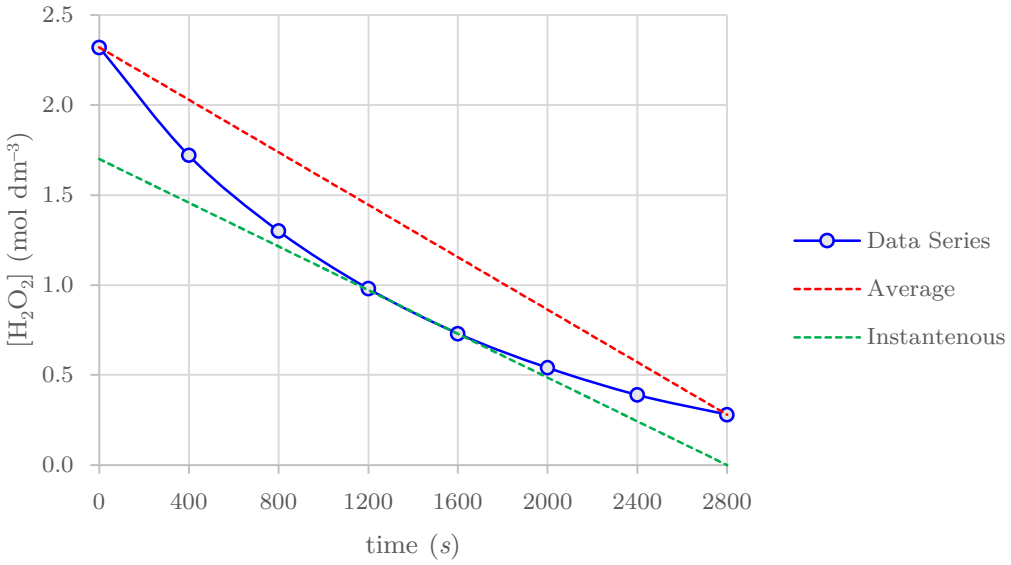

Figure 1: Representation of rate of reaction for the decomposition of $H_2O_2(aq)$. Concentration-time data is typically plotted in this standard format, where concentration is represented over time. The slopes of the tangent lines on the graph are used to calculate the reaction rates.

Time, s	$\Delta t = t_{n+1} - t_n$	$[H_2O_2]$, mol dm^{-3}	$\Delta[H_2O_2]$	r, mol dm^{-3} s^{-1}
0	N.A.	2.32	N.A.	N.A.
400	400	1.72	-0.60	15.0×10^{-4}
800	400	1.30	-0.42	10.5×10^{-4}
1200	400	0.98	-0.32	8.0×10^{-4}
1600	400	0.73	-0.25	6.3×10^{-4}
2000	400	0.54	-0.19	4.8×10^{-4}
2400	400	0.39	-0.15	3.8×10^{-4}
2800	400	0.28	-0.11	2.8×10^{-4}

Table 1: Collected data for the decomposition of $H_2O_2(aq)$.

Rate Law

In a chemical kinetics study, a key objective is to establish an equation that can effectively predict the relationship between the rate of reaction and the concentrations of the reactants involved. This equation, derived through experimental observations, is commonly referred to as a rate law or rate equation.

Consider the following hypothetical reaction,

$$c_a\mathrm{A} + c_b\mathrm{B} \longrightarrow c_x\mathrm{X} + c_y\mathrm{Y}$$

where c_a, c_b, c_x, and c_y are reaction coefficient for each respective chemical species. The rate law expression for the given hypothetical reaction is denotes by,

$$r = k[\mathrm{A}]^{q_a}[\mathrm{B}]^{q_b}[\mathrm{X}]^{q_x}[\mathrm{Y}]^{q_y}$$

where q_a, q_b, q_x, and q_y are order of reaction for each respective chemical species while k is defined as rate constant. The determination of the reaction order, represented by q, cannot be solely based on the reaction coefficients. Experimental investigations are required to calculate the reaction order accurately, with a few exceptions where the reaction follows the law of mass action. In instances where the reaction obeys the law of mass action, the reaction order for each reactant is equal to its respective coefficient. However, in general, experimental data and analysis are necessary to determine the reaction order reliably.

Method of Initial Rates

The method of initial rates is a practical approach in chemical kinetics used to determine the reaction order and rate constant. By conducting a series of experiments with varying initial concentrations of reactants while keeping other conditions constant, the initial rates of reaction are measured. Comparing these initial rates allows for the determination of the reaction order, which represents the exponent or power in the rate equation for each reactant. The reaction order indicates how changes in the concentration of a reactant affect the rate of the reaction.

Consider the reaction for synthesis of $\mathrm{NOCl}(g)$ from the reaction between $\mathrm{NO}(g)$ and $\mathrm{Cl_2}(g)$.

$$\mathrm{NO}(g) + \mathrm{Cl_2}(g) \longrightarrow \mathrm{NOCl}(g)$$

The data for the concentration of $\mathrm{NO}(g)$ and $\mathrm{Cl_2}(g)$, as well as the corresponding initial rates, are presented in Table 2.

Entry	$[\mathrm{NO}]$, mol dm^{-3}	$[\mathrm{Cl_2}]$, mol dm^{-3}	r, mol dm^{-3} s^{-1}
1	0.125	0.255	2.77×10^{-5}
2	0.125	0.510	4.55×10^{-5}
3	0.250	0.255	9.08×10^{-5}

Table 2: Collected data for the synthesis of $\mathrm{NOCl}(g)$ experiment, including measurements of initial reaction rates and determination of reaction orders for each chemical species.

The tentative rate law for the given reaction is expressed as follows:

$$r = k[\mathrm{NO}]^x[\mathrm{Cl_2}]^y$$

To determine the value of x in the rate law expression, we will compare the data from experiment 1 and experiment 3. By examining the changes in the concentration of $\mathrm{NO}(g)$ and their corresponding effect on the rate of reaction, we can establish the reaction order for reactant $\mathrm{NO}(g)$.

$$
\begin{aligned}
\frac{r_1}{r_3} &= \frac{k[\mathrm{NO}]_1^x[\mathrm{Cl_2}]_1^y}{k[\mathrm{NO}]_3^x[\mathrm{Cl_2}]_{23}^y} \\
\frac{2.77 \times 10^{-5}}{9.08 \times 10^{-5}} &= \frac{k(0.125)^x(0.255)^y}{k(0.250)^x(0.255)^y} \\
\frac{277}{908} &= \left(\frac{0.125}{0.250}\right)^x \\
x &\approx 1.712
\end{aligned}
$$

Similarly, to determine the value of y, we will compare the data from experiment 1 and experiment 2. By analyzing the changes in the concentration of $\mathrm{Cl_2}(g)$ and their impact on the rate of reaction, we can establish the reaction order for reactant $\mathrm{Cl_2}(g)$.

$$
\begin{aligned}
\frac{r_1}{r_2} &= \frac{k[\mathrm{NO}]_1^x[\mathrm{Cl_2}]_1^y}{k[\mathrm{NO}]_2^x[\mathrm{Cl_2}]_2^y} \\
\frac{2.77 \times 10^{-5}}{4.55 \times 10^{-5}} &= \frac{k(0.125)^x(0.255)^y}{k(0.125)^x(0.510)^y} \\
\frac{277}{455} &= \left(\frac{0.255}{0.510}\right)^y \\
y &\approx 0.716
\end{aligned}
$$

After determining the values of x and y through the comparison of experimental data, we can incorporate these values into the tentative expression of the rate law. The modified rate law would then be:

$$r = k[\mathrm{NO}]^2[\mathrm{Cl_2}]$$

The rate constant, denoted as k, remains the same and is typically determined separately through experimental measurements.

Integrated Rate Law

The integrated rate law is a mathematical expression that relates the concentration of reactants or products to time during a chemical reaction. It is derived by integrating the rate equation, which describes how the rate of the reaction depends on the concentrations of the reactants. The general form of the integrated rate law can be obtained by considering the differential form of the rate equation and integrating it with respect to time. This resulting equation provides a relationship between the concentration of a reactant or product at a given time and its initial concentration, as well as the rate constant of the reaction. The specific form of the integrated rate law depends on the order of the reaction, which refers to the dependence of the reaction rate on the concentrations of the reactants.

Zeroth Order

In a zeroth order reaction, the rate of the reaction is not influenced by the concentration of the reactants. This type of reaction proceeds at a constant rate, regardless of the initial concentration of the reactants. Examples of zeroth order reactions include certain enzyme-catalyzed reactions or reactions where the rate is limited by the availability of a catalyst or a surface for the reaction to

occur. In these cases, increasing the concentration of the reactants does not affect the reaction rate, as it is already at its maximum.

The concentration of the chemical species, [A], involved in a zeroth order reaction can be determined using the following formulation after a certain time interval Δt:

$$-\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{d}t} = k[\mathrm{A}]^0$$

The exact equation for the differential equation above is,

$$\begin{aligned} k\mathrm{d}t + \mathrm{d}[\mathrm{A}] &= 0 \\ \int_t^{t+\Delta t} k\,\mathrm{d}t &= -\int_{[\mathrm{A}]_t}^{[\mathrm{A}]_{t+\Delta t}} \mathrm{d}[\mathrm{A}] \\ [\mathrm{A}]_{t+\Delta t} &= [\mathrm{A}]_t - k\Delta t \end{aligned}$$

First Order

In a first-order reaction, the rate of the reaction is directly proportional to the concentration of a single reactant. This means that the reaction rate increases or decreases exponentially with changes in the concentration of the reactant. The integrated rate law for a first-order reaction can be derived as follows:

Consider a first-order reaction with reactant A:

$$\mathrm{A} \longrightarrow \mathrm{Product}$$

The rate equation for this reaction can be expressed as:

$$r = k[\mathrm{A}]$$

Where [A] represents the concentration of reactant A and k is the rate constant. To obtain the integrated rate law, we can rearrange this equation and separate the variables:

$$\begin{aligned} -\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{d}t} &= k[\mathrm{A}] \\ \frac{\mathrm{d}[\mathrm{A}]}{[\mathrm{A}]} &= -k\mathrm{d}t \end{aligned}$$

Integrating both sides of the equation gives:

$$\begin{aligned} \int_{[\mathrm{A}]_t}^{[\mathrm{A}]_{t+\Delta t}} \frac{\mathrm{d}[\mathrm{A}]}{[\mathrm{A}]} &= -\int_t^{t+\Delta t} k\,\mathrm{d}t \\ \ln[\mathrm{A}]_{t+\Delta t} &= -k\Delta t + \ln[\mathrm{A}]_t \end{aligned}$$

This equation represents the integrated rate law for a first-order reaction. It relates the natural logarithm of the concentration of reactant A at a given time to the reaction time (Δt) and the rate constant (k). By plotting ln[A] against time, a straight line with a gradient (m) of $-k$ can be observed.

$$\underbrace{\ln[\mathrm{A}]_{t+\Delta t}}_{y} = \underbrace{-k}_{m} \cdot \underbrace{\Delta t}_{x} + \underbrace{\ln[\mathrm{A}]_t}_{c}$$

Second Order

In a second-order reaction, the rate of the reaction is dependent on the concentration of one reactant raised to the power of 2 (squared) or the combined concentrations of two reactants. The integrated rate law for a second-order reaction can be derived as follows:

Consider a second-order reaction with reactants A and B (bimolecular reaction):

$$\mathrm{A} + \mathrm{B} \longrightarrow \mathrm{Product}$$

The rate equation for this reaction can be expressed as:

$$r = k[\mathrm{A}]_t[\mathrm{B}]_t$$

To express the concentrations of reactants A and B at time t, $[\mathrm{A}]_t$ and $[\mathrm{B}]_t$, in terms of changes in concentration, we can consider their initial concentrations $[\mathrm{A}]_0$ and $[\mathrm{B}]_0$. The expressions for $[\mathrm{A}]_t$ and $[\mathrm{B}]_t$ can be formulated as follows:

$$[\mathrm{A}]_t = [\mathrm{A}]_0 - \Delta[\mathrm{A}]$$

$$[\mathrm{B}]_t = [\mathrm{B}]_0 - \Delta[\mathrm{B}]$$

In these expressions, $\Delta[\mathrm{A}]$ represents the change in concentration of reactant A from its initial concentration $[\mathrm{A}]_0$ to its concentration at time t, $[\mathrm{A}]_t$. Similarly, $\Delta[\mathrm{B}]$ represents the change in concentration of reactant B from its initial concentration $[\mathrm{B}]_0$ to its concentration at time t, $[\mathrm{B}]_t$.

If the change in concentration for reactants A and B is equal and denoted as x ($x = \Delta[\mathrm{A}] = \Delta[\mathrm{B}]$), the expression of the rate law can be written as follows:

$$r = k([\mathrm{A}]_0 - x)([\mathrm{B}]_0 - x)$$

To obtain the integrated rate law, we can rearrange this equation and separate the variables:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = k([\mathrm{A}]_0 - x)([\mathrm{B}]_0 - x)$$

$$\frac{\mathrm{d}x}{([\mathrm{A}]_0 - x)([\mathrm{B}]_0 - x)} = k\mathrm{d}t$$

Integrating both sides of the equation gives:

$$\int_0^x \frac{\mathrm{d}x}{([\mathrm{A}]_0 - x)([\mathrm{B}]_0 - x)} = \int_0^t k\,\mathrm{d}t$$

$$\frac{1}{[\mathrm{B}]_0 - [\mathrm{A}]_0}\int_0^x \left(\frac{1}{[\mathrm{A}]_0 - x} - \frac{1}{[\mathrm{B}]_0 - x}\right)\mathrm{d}x = kt$$

$$\frac{1}{[\mathrm{B}]_0 - [\mathrm{A}]_0}\left[\ln\left(\frac{[\mathrm{B}]_0 - x}{[\mathrm{A}]_0 - x}\right)\right]_x^0 = kt$$

$$\frac{1}{[\mathrm{B}]_0 - [\mathrm{A}]_0}\ln\left(\frac{[\mathrm{A}]_0[\mathrm{B}]_t}{[\mathrm{A}]_t[\mathrm{B}]_0}\right) = kt$$

Incase where $[\mathrm{A}]_0 = [\mathrm{B}]_0$ or the reaction is a unimolecular reaction ($\mathrm{A} \longrightarrow \mathrm{Product}$),

$$
\begin{aligned}
kt &= \lim_{[\mathrm{B}]_0 \to [\mathrm{A}]_0} \frac{1}{[\mathrm{B}]_0 - [\mathrm{A}]_0} \ln\left(\frac{[\mathrm{A}]_0[\mathrm{B}]_t}{[\mathrm{A}]_t[\mathrm{B}]_0}\right) \\
&= \lim_{[\mathrm{B}]_0 \to [\mathrm{A}]_0} \frac{1}{[\mathrm{B}]_0 - [\mathrm{A}]_0} \ln\left(\frac{[\mathrm{A}]_0([\mathrm{B}]_0 - x)}{[\mathrm{B}]_0([\mathrm{A}]_0 - x)}\right) \\
&= \frac{x}{[\mathrm{A}]_0([\mathrm{A}]_0 - x)} \\
&= \frac{[\mathrm{A}]_0 - [\mathrm{A}]_t}{[\mathrm{A}]_0[\mathrm{A}]_t}
\end{aligned}
$$

Rearrange the equation,

$$
\begin{aligned}
\frac{1}{[\mathrm{A}]_t} - \frac{1}{[\mathrm{A}]_0} &= kt \\
\frac{1}{[\mathrm{A}]_t} &= \frac{1}{[\mathrm{A}]_0} + kt
\end{aligned}
$$

These equations relate the reciprocal of the concentration of reactant A (or reactants A and B) at a given time to the reaction time (t) and the rate constant (k). By plotting the reciprocal of the concentration against time, a straight line with a slope of k can be observed.

Exercise

Given the bimolecular second order reaction $\mathrm{A} + \mathrm{B} \longrightarrow \mathrm{Product}$. Suppose that the concentration of compound B is in excess amount compared to compound A ($[\mathrm{B}]_0 \gg [\mathrm{A}]_0$).

[a] Show that the rate law for this reaction first order with respect to A (pseudo-first order).

As $[\mathrm{B}]_0 - x \approx [\mathrm{B}]_0$, therefore,

$$r = k([\mathrm{A}]_0 - x)[\mathrm{B}]_0$$

in this case, $[\mathrm{B}]_0$ is taken as constant with respect to x.

[b] Show that the concentration of A after time of Δt can be approximated using the first-order intergrated rate law.

$$\ln[\mathrm{A}]_{t+\Delta t} = -k\Delta t + \ln[\mathrm{A}]_t$$

Models for Chemical Kinetics: Theoretical Approach

The practical aspects of reaction kinetics involve describing rate laws and rate constants without directly considering the behavior of individual molecules. However, gaining a deeper understanding of the underlying processes necessitates examining the molecular-level behavior.

Collision Theory

Collision theory is a cornerstone of chemical kinetics, offering a formal framework for understanding the mechanisms underlying chemical reactions at the molecular level. The theory postulates that for a reaction to occur, reactant molecules must collide with each other in an effective manner. It introduces several key concepts for chemical kinetics:

1. Collision Frequency

The rate of collisions between reactant molecules is directly proportional to their concentrations. This implies that higher concentrations lead to a greater number of collisions per unit time, thereby increasing the chances of a reaction.

2. Activation Energy

   Not all collisions result in a reaction. Only collisions that possess sufficient energy, equal to or exceeding the activation energy (E_a), have the potential to induce the necessary rearrangements of bonds for the reaction to proceed. The activation energy represents the minimum energy required for reactant molecules to overcome the energy barrier and initiate the reaction.

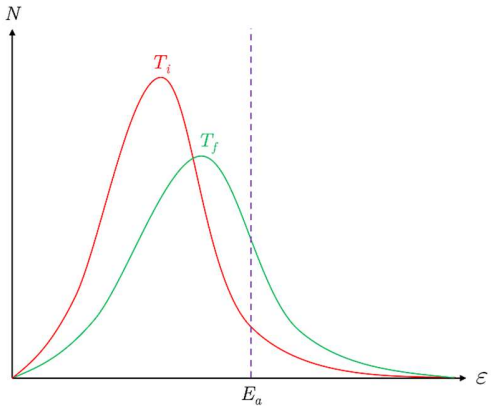

Figure 2: The Boltzmann distribution law describes the distribution of molecules with energies exceeding the activation energy at different temperatures. The graph illustrates the number of molecules (N) having energies (ε) or higher as a function of energy. As the temperature increase ($T_i \rightarrow T_f$), the probability of molecules possessing energies above the activation energy increase exponentially according to the Boltzmann distribution.

Distribution of molecules that exceed the activation energy at different temperatures is described by the Boltzmann distribution law. This law can be formalized as follows:

$$p(i) = \frac{\exp(-\varepsilon_i \beta)}{\sum_{i=1}^{N} \exp(-\varepsilon_i \beta)}; \beta = \frac{1}{k_B T}$$

3. Orientation

   In addition to sufficient energy, collisions must occur with the correct spatial orientation for successful reaction outcomes. The relative orientations of the colliding molecules influence the probability of bond breakage and formation. Proper alignment facilitates the necessary bond rearrangements, while unfavorable orientations may hinder the reaction.

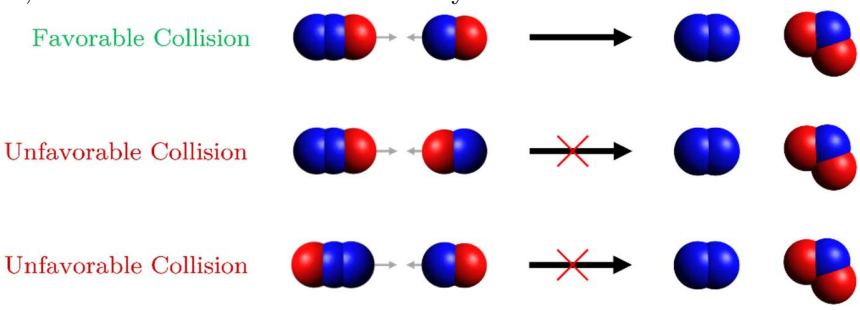

Figure 3: A successful collision between N_2 and NO molecules, leading to the formation of products N_2O and NO_2. Two unsuccessful collisions between N_2 and NO molecules, where no reaction occurs after the collisions.

Nevertheless, it is crucial to acknowledge that collision theory oversimplifies the intricate nature of reactions by disregarding significant factors such as steric effects and specific reaction mechanisms.

Transition State Theory

Henry Eyring (1901–1981) and other theorists introduced a concept that focuses on a hypothetical intermediate state, referred to as the transition state, which lies between the reactants and the products. This transitional state is believed to exist temporarily during a chemical reaction. It is also known as the activated complex. The activated complex, formed through collisions between reactant molecules, can either revert back to the original reactants or proceed to form product molecules.

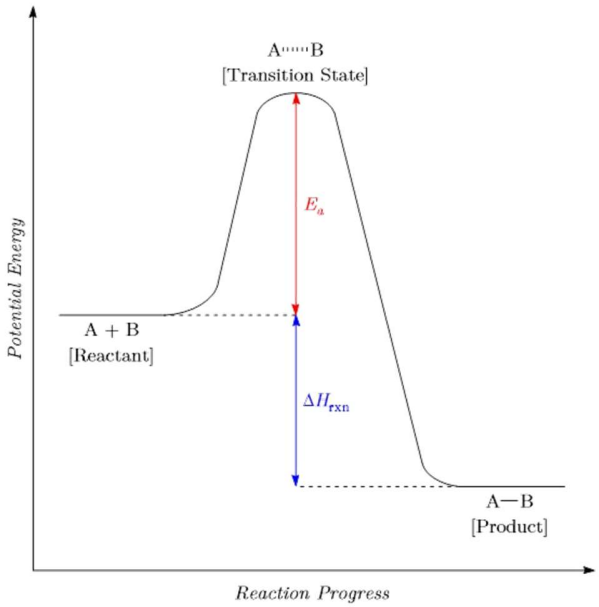

Figure 4: The simplified reaction profile illustrates the variation in energy throughout the course of a reaction. The reactant and product molecules are represented using molecular models, along with the depiction of the activated complex.

Impact of Temperature on Reaction Rates – Arrhenius Equation

The relationship between temperature and reaction rate is typically described mathematically by utilizing two parameters. These parameters correspond to the intercept and slope of a linear plot known as the Arrhenius plot, where the natural logarithm of the reaction rate ($\ln k$) is plotted against the reciprocal of temperature. This mathematical representation is referred to as the Arrhenius equation.

$$\ln k = \ln A - \frac{E_a}{RT}$$

The Arrhenius equation provides two important parameters: A and E_a. The parameter A, obtained from the intercept of the line at ${}^{1}/_{T} \to 0$ (representing infinite temperature), is known as the frequency factor or pre-exponential factor. The parameter E_a, obtained from the slope of the line, is referred to as the activation energy. Together, A and E_a are known as the Arrhenius parameters.

Extrapolation

The rate of the second-order decomposition of ethanal (acetaldehyde, CH_3CHO) was measured over the temperature range 700–1000 K, and the rate constants are reported below.

T (K)	700	730	760	790	810	840	910	1000
k_{rxn}	0.011	0.035	0.105	0.343	0.789	2.170	20.0	145

Table 3: Data tabulation of the rate constant with respect to various temperature.

Based on Arrhenius equation, the data can be analyzed by plotting the natural logarithm of the reaction rate (k/dm^3 mol$^{-1}s^{-1}$) against the reciprocal of temperature or more conveniently, ${}^{10^3}/_{T}$, with the expectation of obtaining a linear relationship. The activation energy can be determined by calculating the dimensionless slope and then using the equation ${}^{\text{slope}}/_{\text{units}} = -{}^{E_a}/_{R}$, where in this case, "units" represents ${}^{1}/_{10^3}$ K^{-1}. Therefore, the activation energy (E_a) can be obtained by multiplying the slope by the gas constant (R) and 10^3 K.

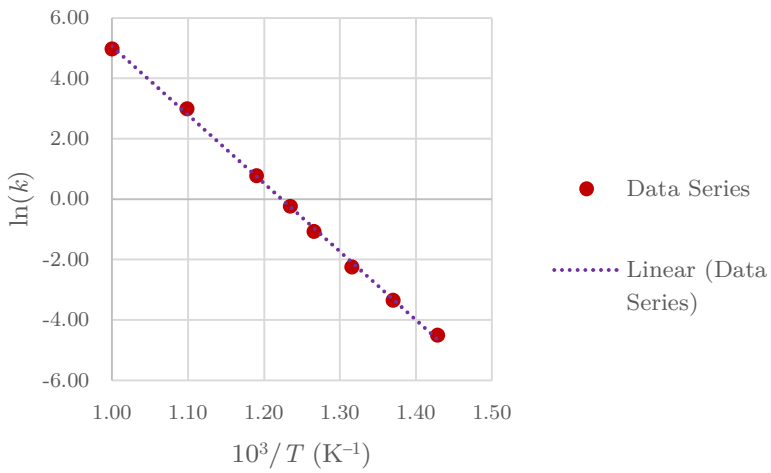

Figure 5: The data was analyzed by extrapolating the natural logarithm of the rate constant as a function of the reciprocal of the temperature. A regression line was used to assess the linearity of the data and determine if there is a consistent relationship between the rate constant and the temperature.

The regression analysis conducted on the data series in the plot above yields the following equation:

$$y = \underbrace{-22.651}_{m}\,x + \underbrace{27.707}_{c}$$

Consequently, the analysis of the regression equation allows us to determine the activation energy (E_a) and the frequency factor (A) for the decomposition of ethanal.

$$E_a = -mR \cdot 10^3 \rightarrow E_a = 188.32 \text{ kJ mol}^{-1}$$

$$\ln A = c \rightarrow A = 1.076 \times 10^{12} \frac{\text{dm}^3}{\text{mol } s}$$

Comparison of Rate Constant at Different Temperature

Consider two temperature T_i and T_f with its corresponding rate constant k_i and k_f. The Arrhenius expression for k_i and k_f is given by,

$$\ln k_i = \ln A - \frac{E_a}{RT_i}$$
$$\ln k_f = \ln A - \frac{E_a}{RT_f}$$

The distinction between the two conditions can be expressed as follows:

$$\ln k_f - \ln k_i = \frac{E_a}{RT_i} - \frac{E_a}{RT_f}$$
$$\ln\left(\frac{k_f}{k_i}\right) = \frac{E_a}{R}\left(\frac{1}{T_i} - \frac{1}{T_f}\right)$$

The relationship between the activation energy and the slope of the plot of $\ln k_r$ against ${}^{1}/_{T}$ leads to the following conclusions:

→ A high activation energy indicates that the rate constant is highly dependent on temperature. In other words, small changes in temperature can have a significant impact on the rate of the reaction. This suggests that the reaction is more sensitive to temperature variations, and even a slight increase in temperature can lead to a substantial increase in the reaction rate.

→ If a reaction has zero activation energy ($E_a = 0$), it implies that the reaction rate is independent of temperature. In such cases, the rate constant remains constant regardless of changes in temperature. This means that the reaction proceeds at the same rate regardless of whether the temperature is increased or decreased. The absence of activation energy suggests that the reaction does not require any additional energy to overcome an energy barrier, resulting in a temperature-independent reaction rate.

→ A negative activation energy ($E_a < 0$) suggests that the reaction rate decreases as the temperature increases. In such cases, as the temperature is raised, the rate constant decreases, resulting in a slower reaction rate. This behavior is contrary to the typical expectation, where higher temperatures usually lead to faster reaction rates. However, in reactions with negative activation energy, the decrease in the rate constant with increasing temperature indicates that the reaction becomes less favorable or encounters some inhibitory factors as the temperature rises.

For certain reactions, it is observed that the plot of $\ln k_r$ against ${}^{1}/_{T}$ does not result in a straight line. However, it is still possible to define an activation energy for these "non-Arrhenius reactions" at a specific temperature by employing the following equation:

$$-\frac{E_a}{R} = \frac{\mathrm{d}\ln k_r}{\mathrm{d}({}^{1}/_{T})} \rightarrow E_a = RT^2\frac{\mathrm{d}\ln k_r}{\mathrm{d}T}$$

The given expression serves as the formal definition of activation energy, which encompasses both temperature-dependent and temperature-independent cases. When the activation energy is independent of temperature, the expression simplifies to the earlier definition, where the activation energy is represented by the slope of a straight line in the Arrhenius plot. However, when a reaction

exhibits non-Arrhenius behavior, meaning that the plot of $\ln k_r$ against $^1/_T$ deviates from a straight line, it suggests that quantum mechanical tunnelling may have a significant influence on the reaction. Quantum tunnelling allows particles to traverse energy barriers that would be classically inaccessible, and its involvement becomes more pronounced in reactions with lower activation energies or particular molecular configurations. Therefore, non-Arrhenius behavior can serve as an indication that quantum tunnelling plays a notable role in the reaction kinetics.

Reaction Mechanism and Molecularity

A reaction mechanism refers to a detailed description of a chemical reaction, outlining the step-by-step process involved. Each individual step within the mechanism is known as an elementary process, which encompasses molecular events that result in substantial changes to a molecule's energy, geometry, or the formation of new molecules. Plausible reaction mechanisms must satisfy two key requirements, which are:

1. Consistency with Stoichiometry
   The mechanism should align with the stoichiometry of the overall reaction. This means that the sum of the elementary steps should yield the correct overall balanced equation for the reaction.
2. Agreement with Experimentally Determined Rate Law
   The mechanism should be able to explain and reproduce the experimentally determined rate law for the reaction. The rate law describes the relationship between the concentrations of reactants and the rate at which the reaction proceeds.

Elementary Processes

The characteristics of elementary processes can be summarized as follows:

1. Unimolecular or Bimolecular
   Elementary processes can either involve the dissociation of a single molecule (unimolecular) or the collision of two molecules (bimolecular). Termolecular processes involving the simultaneous collision of three molecules are relatively rare as elementary processes.
2. Rate Law Exponents
   The exponents of the concentration terms in the rate law for an elementary process are equal to the stoichiometric coefficients in the balanced equation for that process. This is in contrast to the overall rate law, where the exponents may not be directly related to the stoichiometric coefficients in the overall equation.
3. Reversibility and Equilibrium
   Elementary processes are reversible, meaning they can proceed in both the forward and reverse directions. Some elementary processes may reach a state of equilibrium where the rates of the forward and reverse processes are equal.
4. Intermediates
   Certain species are produced in one elementary process and consumed in another. In a proposed reaction mechanism, these intermediates should not appear in either the overall chemical equation or the overall rate law.

5. Rate-Determining Step (RDS)
   The rate of a complex reaction is determined by the slowest step, which is referred to as the *rate-determining step* . If all the steps in a complex reaction occur at similar rates, the rate law can be written based on the complete reaction. In this case, the stoichiometric coefficients of the reactants in the complete reaction are considered as the order of the reaction with respect to the corresponding reactant species.

Consecutive Elementary Reaction

Certain reactions follow a consecutive unimolecular pathway, involving the formation of an intermediate species (referred to as I). This intermediate is formed in one step and subsequently reacts in another step to yield the final products.

$$\mathrm{A} \xrightarrow{k} \mathrm{I} \xrightarrow{k'} \mathrm{P}$$

The rate of decomposition of A (formation of I) is,

$$\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{d}t} = -k[\mathrm{A}] \Longrightarrow [\mathrm{A}]_t = [\mathrm{A}]_0 e^{-kt}$$

Consecutively, the intermediate I is formed due to decomposition of A and consumed to form the desired product P,

$$\frac{\mathrm{d}[\mathrm{I}]}{\mathrm{d}t} = k[\mathrm{A}] - k'[\mathrm{I}]$$

$$\frac{\mathrm{d}[\mathrm{I}]}{\mathrm{d}t} + k'[\mathrm{I}] = k[\mathrm{A}]_0 e^{-kt}$$

In order to solve the given differential equation, we will utilize the method of integrating factor.

$$I_F = \exp\left[\int k' \, \mathrm{d}t\right] \to I_F = e^{k't}$$

$$e^{k't}\frac{\mathrm{d}[\mathrm{I}]}{\mathrm{d}t} + k'[\mathrm{I}]e^{k't} = k[\mathrm{A}]_0 e^{(k'-k)t} \Longrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left([\mathrm{I}]e^{k't}\right) = k[\mathrm{A}]_0 e^{(k'-k)t}$$

$$\int_0^{[\mathrm{I}]_t} \mathrm{d}\left([\mathrm{I}]e^{k't}\right) = \int_0^t k[\mathrm{A}]_0 e^{(k'-k)t} \, \mathrm{d}t$$

Solving the integral will yield the expression of $[\mathrm{I}]_t$,

$$[\mathrm{I}]_t e^{k't} = \frac{k[\mathrm{A}]_0\left(e^{(k'-k)t} - 1\right)}{k' - k}$$

$$[\mathrm{I}]_t = \frac{k[\mathrm{A}]_0}{k' - k}\left(e^{-kt} - e^{-k't}\right)$$

To determine the amount of product (P) at any given time (t), we can apply the principle of mass conservation. According to this law, the total mass of the reactants must be equal to the total mass of the products in a chemical reaction.

$$[\mathrm{A}]_t + [\mathrm{I}]_t + [\mathrm{P}]_t = [\mathrm{A}]_0 \Longrightarrow [\mathrm{P}]_t = \left\{1 + \frac{ke^{-k't} - k'e^{-kt}}{k' - k}\right\}[\mathrm{A}]_0$$

In the given reaction mechanism, the concentration of the intermediate species follows a pattern where it initially increases to a maximum value and then decreases to zero. This indicates that the intermediate species is formed during the reaction but eventually gets consumed completely.

Simultaneously, the concentration of the product species starts at zero and gradually increases as the reaction progresses. As the reaction proceeds, the concentration of the product approaches the initial concentration $[A]_0$ of the reactant A. This implies that once all the reactant A has been converted into the product P, the concentration of P reaches its maximum value, which is equal to $[A]_0$.

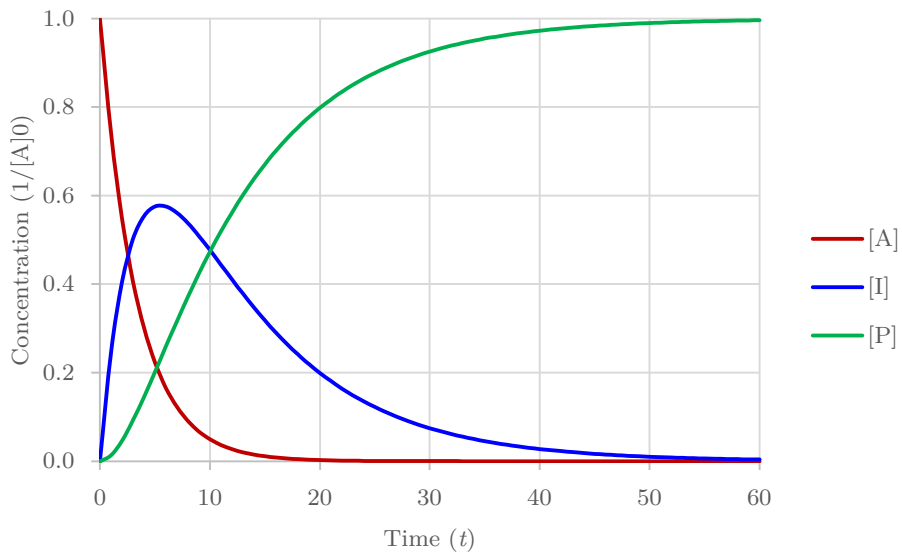

Figure 6: The concentration profiles of species A, I, and P are depicted in the consecutive reaction scheme A⟶I⟶P. The curves represent equations for $[A]_t$, $[I]_t$, and $[P]_t$, where the rate constant k' is three times smaller than the rate constant k ($k = 3k'$). If the intermediate species I is the desired product, it is crucial to be able to anticipate the time at which its concentration reaches its maximum value.

This behavior is consistent with the idea of a complete conversion of reactant A into product P, where the concentration of P gradually increases until all reactant A is consumed.

Quasi Steady State Approximation (QSSA)

The steady state approximation is a mathematical technique used in chemical kinetics to simplify the analysis of reaction mechanisms involving intermediate species. It involves making the following formal assumptions:

[1] Dynamic Equilibrium
The concentration of the intermediate species remains approximately constant over a significant portion of the reaction, establishing a dynamic equilibrium within the system.

[2] Rate of Formation and Consumption
The rate at which the intermediate species is formed is assumed to be equal to its rate of consumption. This assumption implies that the concentration of the intermediate species does not change significantly over time.

[3] Time Derivative of Intermediate Concentration
The time derivative of the concentration of the intermediate species is set to zero in the rate equation, indicating that its concentration remains constant.

To illustrate the utilization of the steady state approximation, let's consider the following decomposition reaction of ozone gas:

$$2O_3(g) \longrightarrow 3O_2(g)$$

The proposed mechanism is

[1] $O_3 \xrightarrow{k_1} O^\bullet + O_2$

[2] $O^\bullet + O_2 \xrightarrow{k_{-1}} O_3$

[3] $O^\bullet + O_3 \xrightarrow{k_2} 2O_2$ [Slow]

For clarity, the first, reversible reaction is written as two separate steps. In order to establish the rate of the overall reaction, we identify a specific step within the reaction mechanism that exhibits a favorable relationship with the experimentally determined rate. In this case, we choose the final step of the mechanism, which encompasses the generation of O_2 and the consumption of the oxygen radical. By focusing on this step, we can establish a connection between the observed rate of the reaction and the concentration of the oxygen radical. This allows us to derive the rate law for the reaction based on this particular step of the mechanism.

$$r = k_2[O^\bullet][O_3]$$

In order to exclude intermediate species from the rate law expression, we can simplify the analysis by assuming that the generation and consumption rates of the oxygen radical ($O^\bullet$) from the equilibrium dissociation of ozone (reaction [1]) and its subsequent consumption reactions (reactions [2] and [3]) are in equilibrium and occur at an equal rate. This assumption, known as the steady-state condition, enables us to express the concentration of the oxygen radical in terms of the concentrations of ozone and oxygen gas.

$$\frac{d[O^\bullet]}{dt} = 0$$

Hence, by considering the equal rates of formation and consumption of the intermediate species (oxygen radical) resulting from the dissociation of ozone and its subsequent reactions, we can express the overall reaction rate as the difference between the rates of formation and consumption of the intermediates.

$$k_1[O_3] - k_{-1}[O^\bullet][O_2] - k_2[O^\bullet][O_3] = 0$$

$$[O^\bullet] = \frac{k_1[O_3]}{k_{-1}[O_2] + k_2[O_3]}$$

Substitute the expression of $[O^\bullet]$ into the rate law,

$$r = \frac{k_1 k_2 [O_3]^2}{k_{-1}[O_2] + k_2[O_3]}$$

Given that reaction [3] is slower in comparison to reactions [1] and [2], we can assume that the rate constant for reaction [2] is much smaller than the rate constants for the reverse reaction [1] ($k_2 \ll k_{-1}$) and the forward reaction [2] ($k_2 \gg k_1$). This simplification allows us to express the rate law in a more straightforward manner, as the contribution of reaction [3] can be considered negligible compared to the other reactions.

$$r \approx \frac{k_1}{k_{-1}k_2}[O_3]^2[O_2]$$

Exercise

Devise the rate law for the decomposition of N_2O_5, $2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$ on the basis of the following mechanism:

[1] $N_2O_5 \xrightarrow{k_1} NO_2 + NO_3$

[2] $NO_2 + NO_3 \xrightarrow{k_{-1}} N_2O_5$

[3] $NO_2 + NO_3 \xrightarrow{k_2} NO_2 + O_2 + NO$

[4] $NO + N_2O_5 \xrightarrow{k_3} 3NO_2$

The rate law for the decomposition of N_2O_5 is given by,

$$\frac{d[N_2O_5]}{dt} = -\frac{2k_1k_2[N_2O_5]}{k_{-1} + k_2}$$